# Logan W.

**Independent project lead**

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I direct AI agents through engineering and research work beyond my own implementation expertise. I set the direction, review the evidence and decide what the record supports; the agents write essentially all of the code. What makes that workable is oversight that doesn't rest on the agents' word: checks that have been seen to fail, verifiers scored against planted faults, and dated records that keep the failed runs.

## Selected projects

### ParcelRound, HonestFramework and HonestHarness

since Sep 2026

- ParcelRound: A method for splitting one body of work across several coding agents at once, without the pieces failing to meet. It exists because the obvious approach fails in a specific, repeatable way: Give four agents four well-scoped parcels of one feature and you can get four green gates and a feature that does not work. 6 rounds are recorded as case studies, and the sixth put planted faults on every parcel's verifier.
- HonestFramework: A way to lay out a project so that an AI can write nearly all of it and no claim about it ever rests on the AI's judgement. Whether a claim is true is answered by something that cannot hold an opinion: a reference implementation, a bit-identity comparison, a hash, a gate that has been watched to fail, an append-only record with dates on it.
- HonestHarness holds the requirements, H1 to H17, derived from the record of cft-fp256's rounds. Its first round asked: can a cheaper model, reached over an API, do the skills a ParcelRound round measures? Replaying the planted faults of ParcelRound's sixth round, DeepSeek's flash model caught 31 of 40, where that round's own verifiers, on Opus, had caught 10 of 10. Every plant that changed a figure or a reference was caught in every run, 24 of 24. Plants that changed meaning or shape were caught 7 of 16.
- Where a suite stops telling strong models apart, the record says so: 659 of 660 runs exact. So the QS6 slice cannot tell strong models apart.
- This site is built the same way. In its second round each verifier, on Haiku, checked a copy of a parcel's work holding planted faults: 8 of 8 caught, each on its first pass.

  Where to look first: ParcelRound's METHOD.md, the case study of HonestHarness's first round.

### cft-fp256

since Aug 2026

- A math coprocessor that gets the same answer everywhere: It does IEEE 754 arithmetic at four precisions - 32, 64, 128 and 256 bits - and guarantees that the same inputs give the same bits whether the work runs on a laptop CPU, in a browser tab, on a microcontroller, or on the FPGA card it was designed for, the AMD/Xilinx Alveo U50 through the open Vitis RTL kernel flow.
- Used cft-fp256 to test how far one person without formal computing training can take a serious engineering effort with AI. The rule is that a number in a document has a run behind it, and the runs that failed stay in the record.
- Checked by 29 simulation targets checking every result and every flag against the golden model, and 38 machine-checked proofs, plus a negative control that must be refuted or the gate has stopped being able to catch a bug; and MPFR arbitrates all four formats.
- The card replayed the conformance sets on silicon: the 1,068,915-case set was replayed on the card, one tile, at each point of the 2026-09-23/24 clock sweep from 145 to 170 MHz. A binary256 soak on the card ran 541 passes in 70 minutes, each a resident run of a million elements at fma, add and mul, with 0 failures.
- cft-rebound is REBOUND's IAS15 integrator with its arithmetic routed through libcft. At binary64 it is REBOUND's own IAS15 bit for bit, which is a gate the port passes on every build.

  Where to look first: the validation ledger, the conformance contract, and the browser replay, which replays the published conformance vectors in front of you.

**Quantum-Film** since Sep 2026

- Film stocks whose crystals are laid by quantum circuits, and a fixer that makes every print of a roll the same, bit for bit, on every machine; its tone curve and grain scale are borrowed from a real stock in atlas-film, so only the placement of the crystals is new.
- Its Pauli stock's law ran on Moth's Atlas emulator, and holds, and the stock was printed, with one print laid entirely by Atlas. On 2026-10-05 the design docs/PREREGISTRATION.md fixed beforehand ran on each of the three IBM Quantum devices the owner's Open-plan instance reaches. The result is reported as it fell: All three devices carry the law's two quantum signatures, well clear of the classical twin. And all three fall short of what their own noise models predicted.
- Winner, Quantum Native Repo category, Moth Hack 2026 (the hackathon's results). Entered for the Expert challenge #9: a repository of a quantum application that runs a process on media through the Atlas API.

**QRO-Crate** since Oct 2026

- An RO-Crate profile for quantum computing runs, built so that the difference between two runs can be attributed: to the question, the compile, what was submitted, the sampling, readout, the device's state and time, or drift within a run. Its first data are Quantum-Film's runs, on IBM's processors and on Moth's Atlas emulator.
- P0 is built, its parts including the canonical writer and the strict reader, the checker's steps 1, 2, 3 and 9 and the custody log and the audit record. Nothing else is built: no adapter, no measures, no comparison, no profile crate, no contributor kit.

## Other public work

- Mercenaries2: A revival project for Mercenaries 2: World in Flames. Restores online services and exposes the game's statically-linked Lua 5.1.2 runtime for modding.
- PrettyCloud, the Atlas of Mathematical Forms: A WebGL2 point-cloud atlas.

## Background

I'm Logan. I have worked as a carpenter since I was fifteen, and I have no formal training in computing. AI agents have written essentially all of the code in these projects; gates decide what is true, and the record keeps what they said.

I have a GED, and one month of formal computer science.

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